Structural Response Theory

A Short Guide

Force-free electrostatics as sources, fields, geometry, and response.

> *This project does not propose a new experimentally distinct force law. It explores a structural reformulation of electrostatic interaction in which force is treated as a derived coordinate-level description rather than a primitive object.*

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Scope: a pedagogical reorganization of classical electrostatics.
No fresh physical content is claimed; Coulomb, Newton, and Maxwell remain correct.

Contents

1 The executive idea

The usual way to teach a charged particle's motion runs

$$\text{source} \longrightarrow \text{force} \longrightarrow \text{motion.}$$

A charge produces a force on a test particle; the force, supplied as a primitive object, then drives Newton's law. This guide reorganizes the same physics along a different chain,

$$\text{source} \longrightarrow \text{structure} \longrightarrow \text{response} \longrightarrow \text{motion,}$$

in which force never appears as an input. Only at the very end, when the response equation is rewritten in coordinates, does the familiar symbol $q\mathbf{E}$ reappear — now as an *output label* rather than a starting point.

Idea

A charge distribution ρ is the **source**. It determines a scalar potential ϕ and the field $\mathbf{E} = -\nabla\phi$, which together form the **structure**. A test particle's trajectory is the **response**: the stationary curve of an action, or the deterministic limit of a noisy drift. The product $q\mathbf{E}$ is what one obtains by writing the response equation $m\ddot{\mathbf{x}} = -q\nabla\phi$ in components. In this telling, "force" is a *derived coordinate-level description*, not the engine of the theory.

Nothing observable changes. Every prediction here is identical to standard electrostatics. The contribution is one of *framing*: a clean source-to-response pipeline, a geometric reading of the inverse-square law, and a small simulation whose internal architecture never treats force as primitive. The remainder of the guide develops each link in the chain and closes with worked exercises.

2 The source → structure → response pipeline

We work throughout in SI units. Write $r = |\mathbf{x}|$ for the distance from the origin and $\hat{r} = \mathbf{x}/r$ for the radial unit vector.

Source. The primitive datum is a charge density $\rho(\mathbf{x})$, a scalar field with dimensions of charge per unit volume. For a single point charge Q at the origin, $\rho = Q\,\delta^3(\mathbf{x})$.

Structure. From ρ we build two objects. The scalar potential ϕ solves the Poisson equation

$$-\Delta\phi = \frac{\rho}{\varepsilon_0}, \tag{1}$$

and the electric field is its (negative) gradient,

$$\mathbf{E} = -\nabla\phi. \tag{2}$$

The pair $(\phi, \mathbf{E})$ is what we call the *structure* induced by the source. It is a property of space, defined everywhere, independent of whether any test particle is present.

Response. A test particle of mass m and charge q moving through this structure follows a trajectory $\mathbf{x}(t)$ determined by a variational principle (Section 5) or, in a dissipative setting, by a stochastic drift (Section 6). The trajectory is the *response*. Reading the response equation in coordinates yields the relation usually written $\mathbf{F} = q\mathbf{E}$.

Note

The order matters pedagogically, not physically. Equations (1) and (2) are exactly Maxwell's electrostatics; the variational law below is exactly Newton's. We are choosing which object to call primitive, and we choose the source and the structure, leaving force to emerge at the end.

3 The Coulomb field from Poisson and the Green function

To turn the source into structure we invert the Poisson operator. The Newtonian Green function for $-\Delta$ in three dimensions is

$$G(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi\,|\mathbf{x} - \mathbf{y}|}, \qquad -\Delta_{\mathbf{x}}\, G(\mathbf{x}, \mathbf{y}) = \delta^3(\mathbf{x} - \mathbf{y}). \tag{3}$$

Convolving the source against the Green function and dividing by ε_0 solves (1):

$$\phi(\mathbf{x}) = \frac{1}{\varepsilon_0} \int G(\mathbf{x}, \mathbf{y})\, \rho(\mathbf{y})\, \mathrm{d}^3 y = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}\, \mathrm{d}^3 y. \tag{4}$$

One checks directly that $-\Delta\phi = \rho/\varepsilon_0$ by passing the Laplacian inside the integral and using (3).

Point charge. For $\rho = Q\,\delta^3(\mathbf{y})$ the integral collapses onto the origin:

$$\phi(\mathbf{x}) = \frac{Q}{4\pi\varepsilon_0}\,\frac{1}{r}, \qquad r = |\mathbf{x}|. \tag{5}$$

The field follows from (2). Using the elementary identity

$$\nabla\left(\frac{1}{r}\right) = -\frac{\mathbf{x}}{r^3} = -\frac{\hat{r}}{r^2}, \tag{6}$$

we obtain

$$\mathbf{E} = -\nabla\phi = -\frac{Q}{4\pi\varepsilon_0}\,\nabla\left(\frac{1}{r}\right) = \frac{Q}{4\pi\varepsilon_0}\,\frac{\hat{r}}{r^2}. \tag{7}$$

This is the Coulomb field, recovered as the structure attached to a point source. It points outward for $Q > 0$ and decays as $1/r^2$. The next section explains *why* the exponent is exactly two.

4 Why $1/r^2$: a flux argument

The inverse-square law is often presented as an axiom. In the structural reading it is a geometric consequence of a conservation statement. Gauss's law, the integral form of (1), states that the flux of $\varepsilon_0\mathbf{E}$ through a closed surface equals the charge it encloses:

$$\oint_{S_r} \varepsilon_0\, \mathbf{E} \cdot \mathbf{n}\, \mathrm{d}S = Q_{\mathrm{enc}}. \tag{8}$$

Take S_r to be a sphere of radius r centered on a point charge Q. By spherical symmetry the field is radial, $\mathbf{E} = E(r)\,\hat{r}$, and the outward normal is $\mathbf{n} = \hat{r}$, so the integrand is constant over S_r. The surface area of the sphere is $4\pi r^2$, hence

$$\varepsilon_0\, E(r)\left(4\pi r^2\right) = Q \qquad \Longrightarrow \qquad E(r) = \frac{Q}{4\pi\varepsilon_0\, r^2}, \tag{9}$$

in agreement with (7).

Note

The $1/r^2$ is **geometric, not an assumed force law**. The enclosed charge Q is fixed, so the total flux through any concentric sphere is the same; only the area over which that fixed flux is distributed grows, and it grows as $4\pi r^2$. The field is flux per unit area, so it must fall as $1/r^2$. The exponent is the dimension-dependent fingerprint of the sphere's area, not an independent postulate about how charges pull on one another.

5 Variational response

With the structure $(\phi, \mathbf{E})$ in hand, the motion of a test particle is fixed by a single scalar functional. Define the action

$$S[\mathbf{x}] = \int \left(\tfrac{1}{2}\, m\, |\dot{\mathbf{x}}|^2 - q\, \phi(\mathbf{x})\right) \mathrm{d}t, \tag{10}$$

the time integral of kinetic minus potential energy, with $L = \frac{1}{2}m|\dot{\mathbf{x}}|^2 - q\phi$. The physical trajectory is the curve that makes S stationary. The Euler–Lagrange equation $\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\mathbf{x}}} = \frac{\partial L}{\partial \mathbf{x}}$ requires the two derivatives

$$\frac{\partial L}{\partial \dot{\mathbf{x}}} = m\dot{\mathbf{x}}, \qquad \frac{\partial L}{\partial \mathbf{x}} = -q\,\nabla\phi, \tag{11}$$

and substituting gives the equation of motion

$$m\ddot{\mathbf{x}} = -q\,\nabla\phi = q\,\mathbf{E}. \tag{12}$$

Equation (12) is the response. Written in coordinates, the right-hand side is the quantity one calls the electric force, $\mathbf{F} = q\mathbf{E}$. But in this derivation it was never supplied: the inputs were the source (through ϕ) and the variational principle. The "force" is the name of $-q\nabla\phi$ once the stationarity condition is expressed component by component — precisely the sense in which it is a derived, coordinate-level description.

6 A note on stochastic response

The variational picture describes conservative motion. Many physical settings instead place the test charge in a viscous medium, where inertia is negligible and the particle drifts along the field while jostled by thermal noise. The natural model is the overdamped Langevin equation

$$\mathrm{d}\mathbf{X} = -\mu\, q\,\nabla\phi(\mathbf{X})\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}, \tag{13}$$

with mobility μ, diffusion constant D, and $\mathbf{W}$ a standard Wiener process. The drift term is again built from the structure ϕ, with no force posited independently. Switching off the noise ($D \to 0$) leaves $\dot{\mathbf{X}} = -\mu q \nabla \phi$, the massless drift; restoring inertia and balancing it against the same gradient returns the deterministic law $m\ddot{\mathbf{x}} = q\mathbf{E}$ of (12).

Note

A stationary probability density of (13) would take the Boltzmann form $p_\infty \propto e^{-\beta q \phi}$ with $\beta = \mu/D$. For this to be normalizable the potential must confine the particle, i.e. $\phi \to +\infty$ at large $|\mathbf{x}|$. The bare Coulomb potential (5) does the opposite — it decays to zero — so no normalizable stationary state exists for a free test charge in an unbounded Coulomb field. A confining term (a trap, a boundary, or an external well) is required before the stationary-state language applies.

7 What is new, and what is not

Honesty is part of the construction. The table separates the standard physics this guide relies on from the modest reframing it offers.

Standard physics (unchanged)	The contribution of this guide
Coulomb's law, Gauss's law, the Poisson equation, and the Newtonian Green function.	A single source-to-response framing that makes the chain explicit and uniform.
The variational (Euler–Lagrange) formulation of mechanics.	A pedagogy in which force is introduced last, as a derived label.
The overdamped Langevin / stochastic-drift description.	A small simulation whose architecture never treats force as a primitive input.

To be unambiguous: there is **no fresh physical content**, **no new force law**, and **no new experimental prediction** anywhere in this document. Newton, Coulomb, and Maxwell remain exactly correct, and every number computed from this framework agrees with the textbook treatment. What is reorganized is the conceptual ordering — source, then structure, then response, with force emerging as the coordinate description of the last step.

8 Exercises with worked solutions

The exercises below trace the pipeline from kernel to interpretation. Each solution is worked in full.

Exercise 1 — Gradient of the Newtonian kernel

For $r = |\mathbf{x}| \neq 0$, compute $\nabla\left(\frac{1}{r}\right)$.

Solution

Start from $r = (x_1^2 + x_2^2 + x_3^2)^{1/2}$, so the i-th partial derivative of r is $\partial_i r = x_i/r$, i.e. $\nabla r = \mathbf{x}/r = \hat{r}$. Apply the chain rule to $f(r) = 1/r$, for which $f'(r) = -1/r^2$:

$$\nabla\left(\frac{1}{r}\right) = f'(r)\,\nabla r = -\frac{1}{r^2}\cdot\frac{\mathbf{x}}{r} = -\frac{\mathbf{x}}{r^3}.$$

$$\boxed{\nabla\left(\frac{1}{r}\right) = -\frac{\mathbf{x}}{r^3} = -\frac{\hat{r}}{r^2}}$$

Exercise 2 — Field from the potential

Given the point-charge potential $\phi = \dfrac{Q}{4\pi\varepsilon_0\, r}$, derive the field $\mathbf{E}$.

Solution

Use $\mathbf{E} = -\nabla\phi$ and pull the constant $Q/(4\pi\varepsilon_0)$ outside the gradient:

$$\mathbf{E} = -\frac{Q}{4\pi\varepsilon_0}\,\nabla\left(\frac{1}{r}\right).$$

Substitute the result of Exercise 1, $\nabla(1/r) = -\mathbf{x}/r^3$:

$$\mathbf{E} = -\frac{Q}{4\pi\varepsilon_0}\left(-\frac{\mathbf{x}}{r^3}\right) = \frac{Q}{4\pi\varepsilon_0}\,\frac{\mathbf{x}}{r^3}.$$

Writing $\mathbf{x}/r^3 = \hat{r}/r^2$ gives the radial form. The field points outward for $Q > 0$, as expected for a positive source.

$$\boxed{\mathbf{E} = \frac{Q}{4\pi\varepsilon_0\, r^2}\,\hat{r}}$$

Exercise 3 — Inverse-square law from flux

Show that Gauss's law together with spherical symmetry forces $E \propto 1/r^2$.

Solution

Gauss's law for a sphere S_r of radius r centered on the charge reads $\oint_{S_r} \varepsilon_0\,\mathbf{E}\cdot\mathbf{n}\,dS = Q_{\text{enc}}$. Spherical symmetry forces the field to be radial with a magnitude depending only on r, $\mathbf{E} = E(r)\,\hat{r}$, and the outward normal is $\mathbf{n} = \hat{r}$, so $\mathbf{E}\cdot\mathbf{n} = E(r)$ is constant over the surface. The surface integral is then $E(r)$ times the area $4\pi r^2$:

$$\varepsilon_0\, E(r)\,(4\pi r^2) = Q \qquad \Longrightarrow \qquad E(r) = \frac{Q}{4\pi\varepsilon_0\, r^2}.$$

The enclosed charge Q is fixed and the total flux is therefore radius-independent; the field falls as $1/r^2$ purely because the same flux is spread over an area that grows as $4\pi r^2$.

$$\boxed{E(r) = \frac{Q}{4\pi\varepsilon_0\, r^2} \propto \frac{1}{r^2}}$$

Exercise 4 — Variational response

Derive the equation of motion from the Lagrangian $L = \frac{1}{2}m|\dot{\mathbf{x}}|^2 - q\,\phi(\mathbf{x})$.

Solution

Compute the two Euler–Lagrange ingredients. The conjugate momentum is

$$\frac{\partial L}{\partial \dot{\mathbf{x}}} = m\dot{\mathbf{x}}, \qquad \text{so} \qquad \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\mathbf{x}}} = m\ddot{\mathbf{x}},$$

and the configuration-space derivative is

$$\frac{\partial L}{\partial \mathbf{x}} = -q\,\nabla\phi.$$

The Euler–Lagrange equation $\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\mathbf{x}}} = \frac{\partial L}{\partial \mathbf{x}}$ equates them:

$$m\ddot{\mathbf{x}} = -q\,\nabla\phi = q\,\mathbf{E}.$$

The trajectory is the stationary curve of the action; the relation usually written $\mathbf{F} = q\mathbf{E}$ is exactly this equation expressed in coordinates.

$$\boxed{m\ddot{\mathbf{x}} = q\,\mathbf{E}}$$

Exercise 5 — Why force is derived, not primitive

Explain in what precise sense force is not a starting object in this construction.

Solution

Trace the inputs at each stage of source $\to$ structure $\to$ response. *(i)* The source is the charge density ρ; it is given. *(ii)* The structure ϕ is computed from ρ by solving $-\Delta\phi = \rho/\varepsilon_0$ (equivalently by the Green convolution (4)), and the field is $\mathbf{E} = -\nabla\phi$ — no force is used. *(iii)* The response is the stationary curve of the action (10) (or the deterministic limit of the drift (13)); the inputs are ϕ and a variational principle, again with no force supplied. The symbol $q\mathbf{E}$ appears only when the response equation $m\ddot{\mathbf{x}} = -q\nabla\phi$ is rewritten component by component. Hence force is an *output label* attached to $-q\nabla\phi$, not a primitive fed into the theory. This is a reorganization of standard electrostatics, not a change to its content.

$$\boxed{\text{Force} = q\mathbf{E} \text{ is the coordinate reading of the response, derived from } (\rho, \phi).}$$

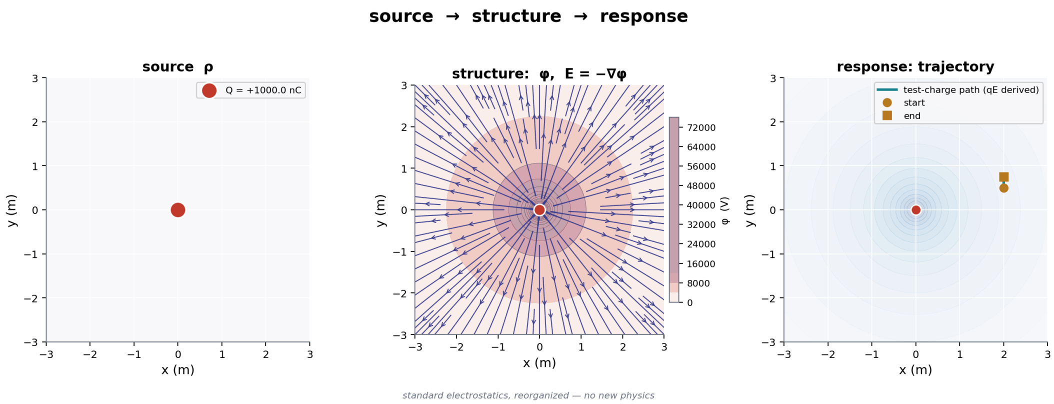

Figure 1: The source $\longrightarrow$ structure $\longrightarrow$ response pipeline.

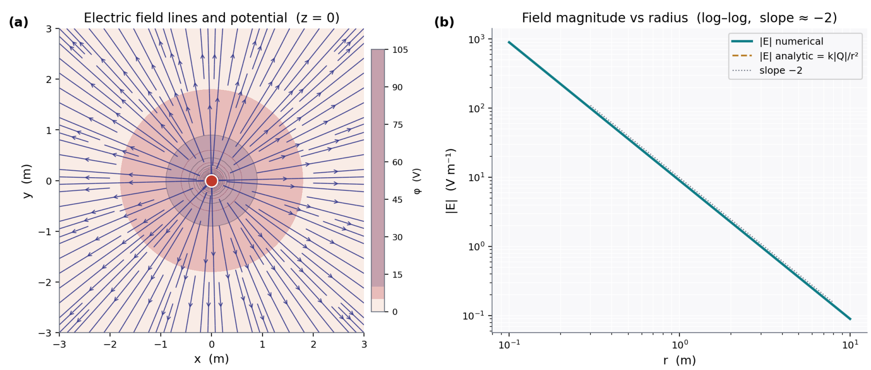

Figure 2: Potential contours and field lines of a point charge.

Exercise 6 — Reading a flux-invariance plot

A numerical experiment evaluates the flux of $\varepsilon_0 \mathbf{E}$ through concentric spheres of increasing radius around a point charge Q and plots the dimensionless ratio Φ/Q against radius. The curve is essentially flat at $\Phi/Q \approx 1$. Interpret it.

Solution

By Gauss's law (8) the flux of $\varepsilon_0 \mathbf{E}$ through any closed surface equals the enclosed charge, so for every sphere around Q the exact value is $\Phi = Q$, i.e. $\Phi/Q = 1$ independent of radius. A flat curve at 1 therefore confirms that the numerical field reproduces this conservation statement across radii; small departures from 1 measure discretization error, not physics. Read together with the area growth $4\pi r^2$, the same flatness is the origin of the inverse-square decay: constant flux divided by an area $\propto r^2$ gives a field $\propto 1/r^2$.

$$\boxed{\Phi/Q \approx 1 \text{ for all } r \iff \text{Gauss's law holds} \implies E \propto 1/r^2.}$$

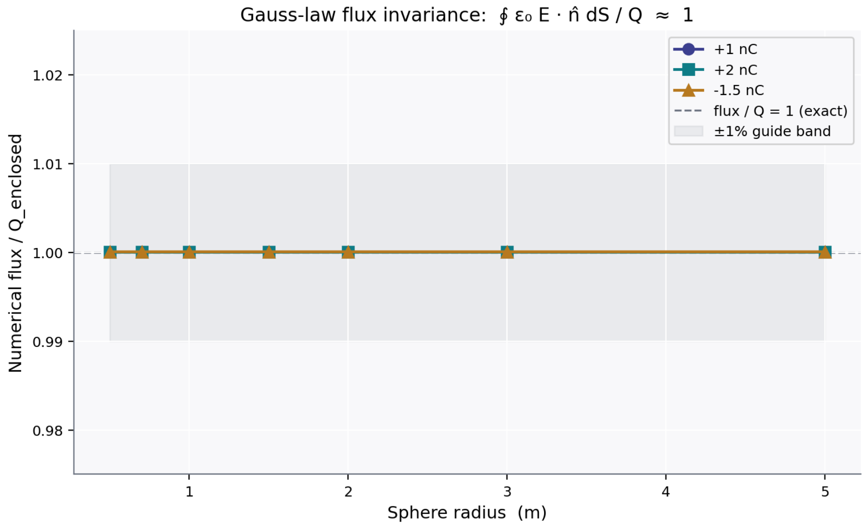

Figure 3: Numerical flux divided by enclosed charge across sphere radii.

Summary. A charge distribution is the source; the potential and field are the structure it induces; a trajectory is the response of a test particle to that structure. The relation $\mathbf{F} = q\mathbf{E}$ *is the coordinate description of the response, recovered at the end rather than assumed at the start. The physics is standard throughout.*